

EDLD_650_dare_test_2

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EDLD 650: Data Analysis and Replication Exercise

(DARE) 2

A. Assumption tests

A1. Describes whether the forcing variable predicts the question variable of interest./

Answer Figure 1 displays the relationship between schools' 2003 Academic Performance Index (API) scores and the probability of receiving additional instructional material funding in 2005, with the forcing variable centered at the eligibility cutoff of 643 (normalized to 0). A clear discontinuity is evident at the threshold, with schools just below the cutoff showing a substantially higher probability of receiving funding, confirming that the forcing variable strongly predicts treatment assignment as intended by the Williams settlement funding policy.

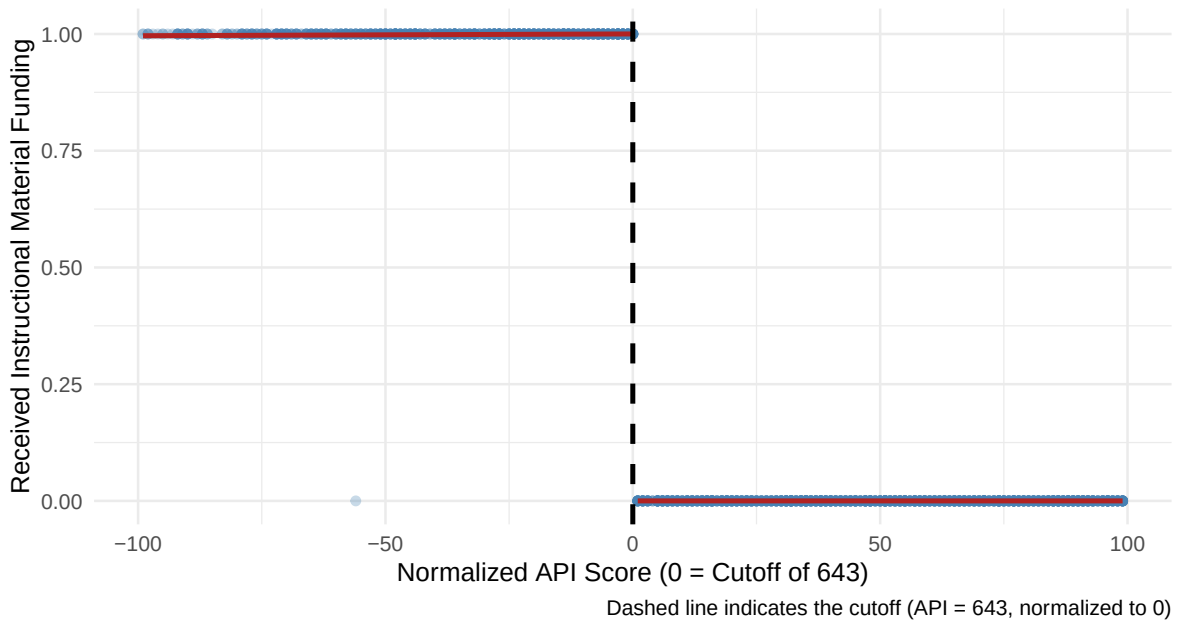
```
data <- read_dta(here("data/EDLD_650_CA_schools_es.dta"))

data_2005 <- data %>%
  filter(year == 2005) %>%
  filter(!is.na(receive_williams), !is.na(norm))

ggplot(data_2005, aes(x = norm, y = receive_williams)) +
  geom_point(alpha = 0.3, size = 1.5, color = "steelblue") +
  geom_smooth(data = filter(data_2005, norm <= 0),
              method = "lm", se = TRUE, color = "firebrick") +
  geom_smooth(data = filter(data_2005, norm > 0),
              method = "lm", se = TRUE, color = "firebrick") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "black", linewidth = 1) +
  labs(
    title = "Figure 1: Effect of API Threshold on Instructional Material Funding (2005)",
    x = "Normalized API Score (0 = Cutoff of 643)",
    y = "Received Instructional Material Funding",
    caption = "Dashed line indicates the cutoff (API = 643, normalized to 0)"
  )
```

```
) +  
theme_minimal()
```

Figure 1: Effect of API Threshold on Instructional Material Funding (2005)



A2. Is there evidence of schools manipulating their placement around the discontinuity?/

Answer Figure 2a shows the distribution of schools' 2003 API scores around the eligibility cutoff. The distribution is smooth with no unusual bunching of schools just below the threshold, suggesting that schools did not manipulate their scores to qualify for Williams funding.

```
data_2003 <- data%>%  
  filter(year == 2003)  
  
ggplot(data_2003, aes(x = norm)) +  
  geom_histogram(binwidth = 5, fill = "steelblue", color = "white", alpha = 0.8) +  
  geom_vline(xintercept = 0, linetype = "dashed", color = "firebrick", linewidth = 1) +  
  labs(  
    title = "Figure 2a: Distribution of Normalized API Scores (2003)",  
    x = "Normalized API Score (0 = Cutoff of 643)",  
    y = "Number of Schools",  
    caption = "Dashed line indicates the eligibility cutoff (API = 643)"  
  ) +  
  theme_minimal()
```

Figure 2a: Distribution of Normalized API Scores (2003)

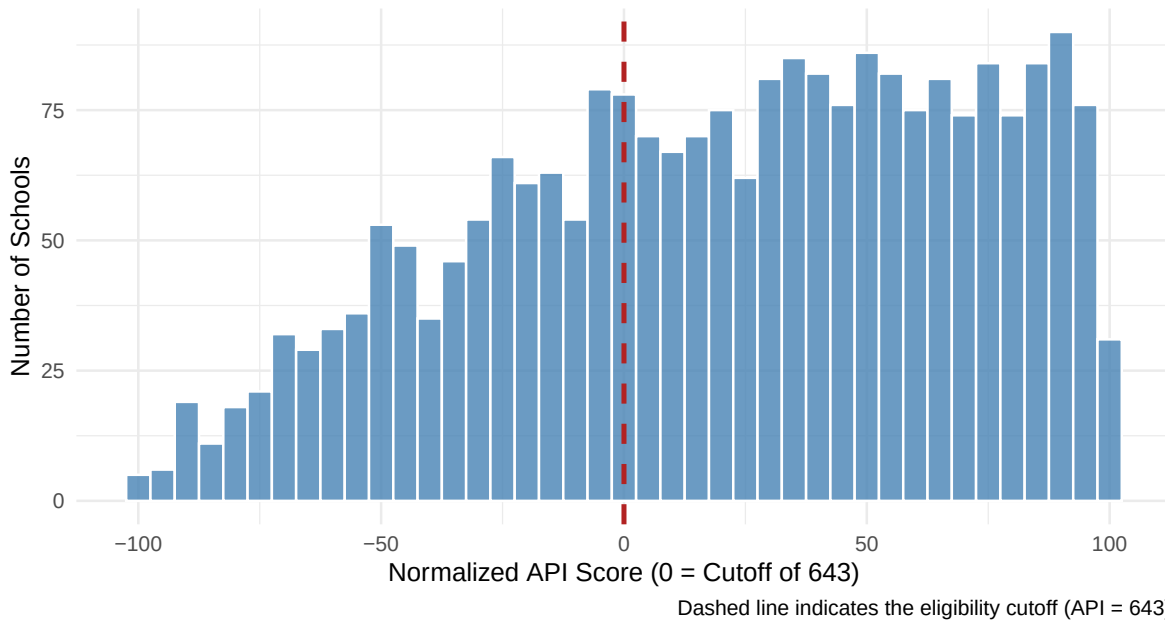


Figure 2b examines whether schools on either side of the cutoff looked similar on key pre-treatment characteristics, including free/reduced lunch eligibility, class size, years of teaching experience, and racial composition, none of which could have been affected by receiving Williams funding. The lack of any sharp jump in these characteristics at the cutoff suggests that schools just below and just above the threshold were comparable, supporting the validity of the regression discontinuity design.

```
data_2003_long <- data_2003 %>%
  mutate(above_cutoff = ifelse(norm > 0, "Above Cutoff", "Below Cutoff")) %>%
  select(norm, above_cutoff, percentfrl, classsize, yrs_teach, pct_hi, pct_aa) %>%
  pivot_longer(cols = c(percentfrl, classsize, yrs_teach, pct_hi, pct_aa),
               names_to = "covariate",
               values_to = "value") %>%
  mutate(covariate = recode(covariate,
    "percentfrl" = "% Free/Reduced Lunch",
    "classsize" = "Class Size",
    "yrs_teach" = "Years Teaching",
    "pct_hi" = "% Hispanic Students",
    "pct_aa" = "% Black Students"
  ))

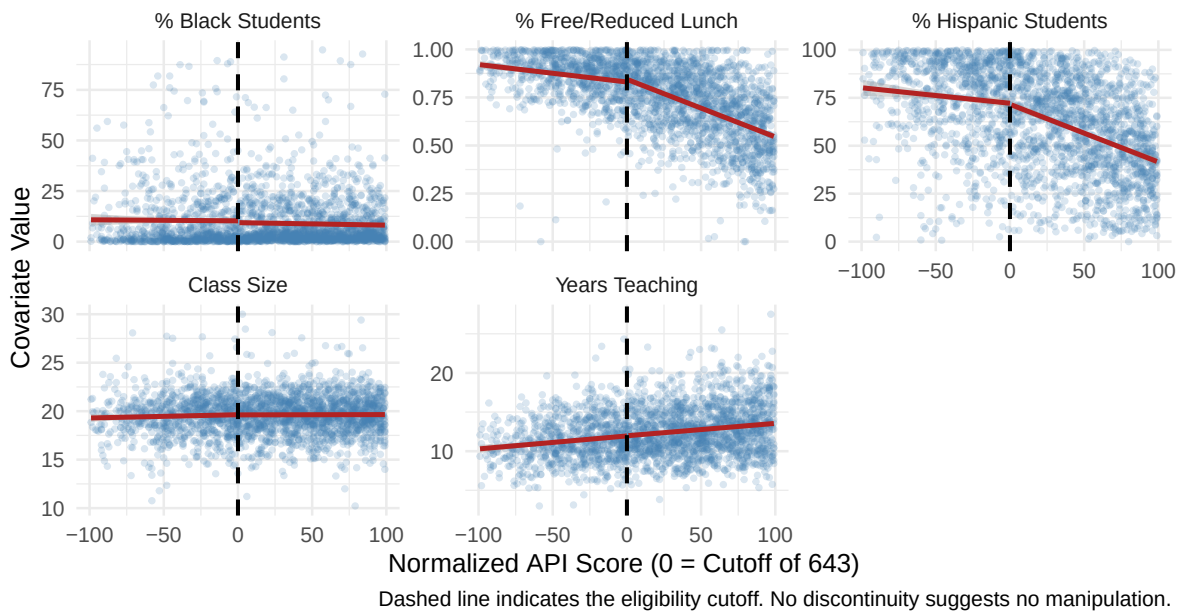
ggplot(data_2003_long, aes(x = norm, y = value)) +
  geom_point(alpha = 0.2, size = 0.8, color = "steelblue") +
```

```

geom_smooth(data = filter(data_2003_long, norm <= 0),
            method = "lm", se = TRUE, color = "firebrick") +
geom_smooth(data = filter(data_2003_long, norm > 0),
            method = "lm", se = TRUE, color = "firebrick") +
geom_vline(xintercept = 0, linetype = "dashed", color = "black", linewidth = 0.8) +
facet_wrap(~ covariate, scales = "free_y") +
labs(
  title = "Figure 2b: Covariate Continuity Around the API Cutoff (2003)",
  x = "Normalized API Score (0 = Cutoff of 643)",
  y = "Covariate Value",
  caption = "Dashed line indicates the eligibility cutoff. No discontinuity suggests no manipulation."
) +
theme_minimal()

```

Figure 2b: Covariate Continuity Around the API Cutoff (2003)



A3. Construct a table of summary statistics./

Answer Table 1 presents summary statistics for the 2,353 elementary schools in our sample in 2003. Compared to Holden (2016)'s full California sample, our data covers fewer years (2003–2009 vs. 2002–2011) and skews toward lower-performing, higher-poverty schools, with a higher share of Hispanic students (62.72% vs. 47.01%) and free/reduced lunch eligibility (75% vs. 54.38%).

Because our sample is concentrated near the eligibility threshold, estimates are most relevant to schools around the cutoff and may not generalize to the broader California elementary school

population , consistent with the local average treatment effect interpretation of regression discontinuity designs.

```
summary_table <- data_2003 %>%
  summarise(
    `Math Score (mean)` = round(mean(mathscore, na.rm = TRUE), 2),
    `Math Score (sd)` = round(sd(mathscore, na.rm = TRUE), 2),
    `Reading Score (mean)` = round(mean(readingscore, na.rm = TRUE), 2),
    `Reading Score (sd)` = round(sd(readingscore, na.rm = TRUE), 2),
    `% Hispanic (mean)` = round(mean(pct_hi, na.rm = TRUE), 2),
    `% Black (mean)` = round(mean(pct_aa, na.rm = TRUE), 2),
    `% White (mean)` = round(mean(pct_wh, na.rm = TRUE), 2),
    `% Free/Reduced Lunch (mean)` = round(mean(percentfrl, na.rm = TRUE), 2),
    `Class Size (mean)` = round(mean(classsize, na.rm = TRUE), 2),
    `Years Teaching (mean)` = round(mean(yrs_teach, na.rm = TRUE), 2),
    `Observations` = n()
  ) %>%
  pivot_longer(everything(), names_to = "Variable", values_to = "Value")

kable(summary_table, caption = "Table 1: Summary Statistics, Elementary Schools (2003)")
```

Table 1: Table 1: Summary Statistics, Elementary Schools (2003)

Variable	Value
Math Score (mean)	324.65
Math Score (sd)	15.30
Reading Score (mean)	316.41
Reading Score (sd)	12.25
% Hispanic (mean)	62.72
% Black (mean)	9.34
% White (mean)	18.64
% Free/Reduced Lunch (mean)	0.75
Class Size (mean)	19.59
Years Teaching (mean)	12.29
Observations	2353.00

B. Replication and Extension

B1. Did the receipt of additional funds for instructional materials improve test score outcomes for elementary students in California? Construct a figure that presents graphical evidence in support of your answer to this question./

Answer In Figure 3, we seek to provide visual evidence of the impact of additional textbook funds for instructional materials on student achievement by comparing the baseline score in 2003 with post-treatment outcomes in 2005. Following Holden (2016), we utilize a bandwidth of 19.099 API points for the main analysis. Panel A of Figure 3 shows school-level test scores in 2003, before the IMWC textbook funding is assigned, and panel B shows school-level test scores in 2005, after the IMWC textbook funding is distributed, both as a function of 2003 API. Visual inspection suggests that test scores are discontinuously higher after IMWC textbook funding is assigned while pre-treatment outcomes are similar.

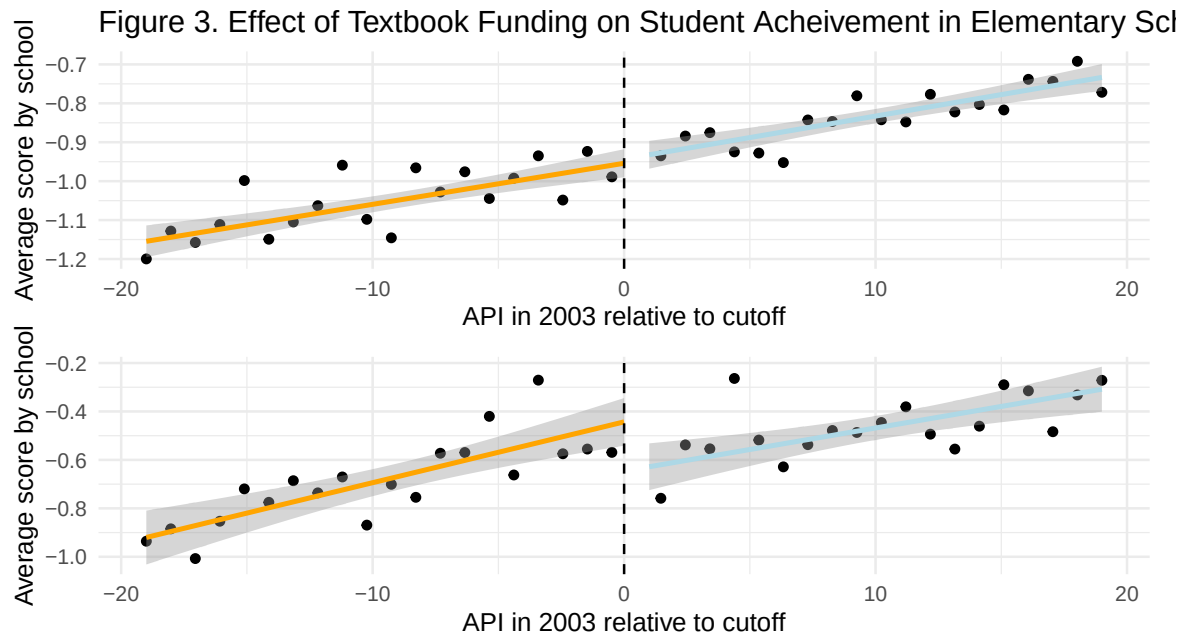
```
data_2003_analysis <- data_2003 %>%
  filter(abs(norm) <= 19.099)

data_2005_analysis <- data_2005 %>%
  filter(abs(norm) <= 19.099)

panel_a <- ggplot(data_2003_analysis , aes(x = norm, y = average_score)) +
  stat_summary_bin(fun = "mean", bins = 40, geom = "point") +
  geom_smooth(data = filter(data_2003_analysis, norm <= 0),
              method = "lm", color = "orange") +
  geom_smooth(data = filter(data_2003_analysis, norm > 0),
              method = "lm", color = "lightblue") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(title = "Figure 3. Effect of Textbook Funding on Student Achievement in Elementary Schools",
       x = "API in 2003 relative to cutoff",
       y = "Average score by school") +
  theme_minimal()

panel_b <- ggplot(data_2005_analysis , aes(x = norm, y = average_score)) +
  stat_summary_bin(fun = "mean", bins = 40, geom = "point") +
  geom_smooth(data = filter(data_2005_analysis, norm <= 0),
              method = "lm", color = "orange") +
  geom_smooth(data = filter(data_2005_analysis , norm > 0),
              method = "lm", color = "lightblue") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(x = "API in 2003 relative to cutoff",
       y = "Average score by school") +
  theme_minimal()

panel_a/panel_b
```



B2. In a regression framework, formally test whether the receipt of additional funds for instructional materials improved test score outcomes for elementary students in California./

Answer In Table 2, we present estimates for the observed discontinuity seen in panel B of Figure 3 by year. Each entry represents a separate regression, where the row identifies the dependent variable measuring student achievement and the column represents the year. We utilize the same bandwidth as Holden(2016), so each entry is an estimated effect from a linear regression with flexible slopes, rectangular kernel weights and a bandwidth of 19.099. Similar to panel A of Figure 3, column 2 to 3 present the estimated discontinuities in test scores as falsification test, while columns 1 and 4 to 8 present estimated program effects. The estimates in column 4 indicate an effect on student achievement of 0.20 standard deviations in the first year that is statistically significant at the 1 percent level of average scores($p < 0.05$). Same as the results being displayed in Holden's analysis (2016), the estimated effect declines to 0.16 and 0.18 standard deviations in 2006 and 2007 and is not significant in later years.

```
data_bandwidth <- data %>%
  filter(abs(norm) <= 19.099)

data_2003 <- subset(data_bandwidth, year == 2003)
data_2004 <- subset(data_bandwidth, year == 2004)
data_2005 <- subset(data_bandwidth, year == 2005)
data_2006 <- subset(data_bandwidth, year == 2006)
data_2007 <- subset(data_bandwidth, year == 2007)
data_2008 <- subset(data_bandwidth, year == 2008)
```

```

data_2009 <- subset(data_bandwidth, year == 2009)
data_all_post <- subset(data_bandwidth, year >= 2005)

fit_all <- lm(average_score ~ ind * norm, data = data_all_post)
fit_2003 <- lm(average_score ~ ind * norm, data = data_2003)
fit_2004 <- lm(average_score ~ ind * norm, data = data_2004)
fit_2005 <- lm(average_score ~ ind * norm, data = data_2005)
fit_2006 <- lm(average_score ~ ind * norm, data = data_2006)
fit_2007 <- lm(average_score ~ ind * norm, data = data_2007)
fit_2008 <- lm(average_score ~ ind * norm, data = data_2008)
fit_2009 <- lm(average_score ~ ind * norm, data = data_2009)

models_avg <- list(
  "All Post" = fit_all,
  "2003"     = fit_2003,
  "2004"     = fit_2004,
  "2005"     = fit_2005,
  "2006"     = fit_2006,
  "2007"     = fit_2007,
  "2008"     = fit_2008,
  "2009"     = fit_2009
)

modelsummary(models_avg,
  vcov = "HC1",
  stars = TRUE,
  coef_map = c("ind" = "Treatment Effect"),
  gof_omit = "AIC|BIC|Log.Lik|RMSE",
  title = "Table 2. Regression Discontinuity Estimates of Effect of Textbook on S",
  notes = "Notes: Each entry is an estimated effect from a linear regression with

```

B3. Write a discussion paragraph in which you present the substantive conclusions (and limitations) of your results about the effects of added textbook funding for California elementary school students.

Answer

The results showing in our estimate of effects of additional textbook funding on school's average test score can help us to understand the effects of added textbook funding for California elementary school students. The significant 0.20 increase in average test scores for 2005 suggests that targeted resources can provide immediate influence to student achievement in elementary schools. However, this magnitude represents only the effect in the first year when eligible schools received the funds, and the observed effect decay in the following years from 2006 to

Table 2: Table 2. Regression Discontinuity Estimates of Effect of Textbook on Student Achievement in Elementary Schools

	All Post	2003	2004	2005	2006	2007	2008	2009
Treatment Effect	0.163*** (0.049)	-0.011 (0.028)	0.095+ (0.055)	0.203** (0.073)	0.164+ (0.087)	0.183* (0.093)	0.117 (0.099)	0.148 (0.105)
Num.Obs.	2680	536	536	536	536	536	536	536
R2	0.047	0.378	0.127	0.119	0.089	0.076	0.045	0.034
R2 Adj.	0.046	0.375	0.122	0.115	0.083	0.071	0.039	0.028
Std.Errors	HC1	HC1	HC1	HC1	HC1	HC1	HC1	HC1

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Each entry is an estimated effect from a linear regression with flexible slopes, rectangular kernel weights and a bandwidth of 19.099. Robust standard errors are displayed in parentheses. Estimates in all post years poll all observations from 2005 to 2009 for a total of 2,680 observations. Columns 2-9 show estimated effects for individual years, with 536 observations each.

2009 suggests that one-time intervention may not be sufficient to sustain long-term academic achievement.

Our identification strategy relies on the assumption that schools that differ in API by only a few points are equal in expectation prior to the exogenous assignment to eligibility of Williams settlement. Our modeling approach depends on our ability to project a smooth relationship between average scores and API across the discontinuity and then estimate the discontinuous effect of being eligible to receive the Williams settlement funds.

Acknowledgement In the process of completing this assignment, we utilized Claude/ChatGpt to clean or debug our code and refine our answers.